Lecture 6.4: Reading a Methods Section

AREC 570 / ECON 530 · Unit 6

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Before class

Read

- Gibson, “Are You Estimating the Right Thing? An Editor Reflects” (Gibson 2019)
- Kim, “Tackling False Positives in Business Research” (Kim 2019)

Optional background for the methods survey

- Falk and Heckman on laboratory experiments (Falk and Heckman 2009)
- Starr on qualitative and mixed methods (Starr 2014)

Readings and course handouts are available on Canvas

How to read a methods section

Follow the argument from the research question to the conclusion

1. What quantity is the study trying to learn about
2. Which observations and measures represent that quantity
3. What comparison or variation supports the estimate
4. How are the estimate and its uncertainty calculated
5. Which claims do the data and assumptions support

Make two passes through the section

Pass	Read for	Mark in the paper
1. Purpose	The estimand, population, and outcome	One sentence stating what is estimated
2. Evidence	Sample, variation, assumptions, and inference	The paragraph or table supporting each claim

Then ask where the paper explains the largest threat to its interpretation

First check: what is being estimated?

An **estimand** names the target quantity

Part	Example for a water-pricing study
Population	Farms served by participating irrigation districts
Outcome	Monthly irrigation water use
Comparison	Use under a revised price and under the prior price
Period	The first two years after the price change

A reviewer looks for a result that answers the question stated in the introduction

Sources: Gibson (2019); Lecture 6.1

Gibson: check whom the data describe

Gibson describes studies of local residents that use administrative records of **registered** residents

Registration and actual residence can differ when people move

Ask which population the data cover, who is missing, and whether the difference matters for the proposed estimate

Source: Gibson (2019), pages 331 to 332

Gibson: check what the measure records

A household survey may record **purchases during one week**. A household can report zero purchases and still consume food from its existing stocks

For every central variable, ask

- What event or quantity is recorded
- What period does it cover
- How does the paper construct it
- Does it represent the concept in the research question

Source: Gibson (2019), discussion of food purchases and consumption

What reviewers check in the data

Item	Question
Sample	How were units selected and which were excluded
Observation	What does one row represent
Measures	How were the outcome and explanatory variables constructed
Timing	Were the relevant measures observed before or after exposure
Missingness	Which observations are absent and why

For survey data, also check the sampling frame, response process, and weights

Sources: Gibson (2019); Lecture 5.1

Methods answer different kinds of questions

Research task	Example of evidence or method
Estimate a policy effect	Experiment or credible observational comparison
Estimate preferences	Choice data with an economic model
Predict a future outcome	Held-out evaluation of a prediction rule
Understand a decision process	Interviews, fieldwork, or case studies
Isolate a proposed mechanism	Controlled laboratory variation

Falk and Heckman: controlled variation

In a gift-exchange experiment, researchers can vary a wage offer and observe effort while specifying the decision rules and information available to participants

When reading an experiment, identify

- What the researcher varies
- What is held fixed
- What participants choose and what is observed
- Which populations and settings the result is meant to inform

Source: Falk and Heckman (2009), discussion of the gift-exchange experiment

Starr: open-ended evidence

Interviews and fieldwork can reveal how people describe a decision and which considerations the researcher had not anticipated

Read for the recruitment rule, interview protocol, record keeping, and method of analyzing responses

Ask how the authors compare accounts with other evidence and address cases that challenge their explanation

Source: Starr (2014), Sections 2 and 3

When methods are combined

Sequence	What the added method contributes
Interviews before a survey	Clarify concepts and improve survey questions
Survey before interviews	Investigate patterns or unexpected responses
Concurrent evidence	Compare accounts with measured outcomes

A methods section should explain how the components inform the same research question

Source: Starr (2014), Section 2

What makes a comparison credible?

For an effect estimate, locate four statements in the paper

1. The groups, places, or periods being compared
2. The source of variation in the explanatory factor
3. The assumption that gives the comparison its interpretation
4. The main ways that assumption could fail

Then examine how the authors investigate those threats

What details permit replication?

Methods detail	What a reader needs
Equation or rule	Outcome, explanatory variables, and target parameter
Sample	Units, period, restrictions, and missing-data rules
Estimation	Estimator, controls, and model settings
Inference	Standard errors, clustering, and tests
Checks	Planned sensitivity and alternative specifications

False positives and false negatives

Decision	Null is true	Specified effect is present
Reject the null	**False positive** or Type I error	Detect the effect
Do not reject	Correct non-rejection	**False negative** or Type II error

The Type I error rate is α. Power is the probability of detecting a specified effect, $1 - \beta$

Source: Kim (2019), Section 2

Power begins with an effect worth detecting

Choose the smallest effect that would matter for the research or policy decision

Power for that effect depends on

- Outcome variation and measurement quality
- The number of usable observations in each group
- Clustering, treatment allocation, and the planned estimator
- The test level and the direction of the test

State the assumed effect size and design when reporting a power calculation

Source: Kim (2019), Sections 2 and 3.5

Survey size has two different uses

Goal	Planning question
Describe a population	How precise must a proportion or mean be
Compare groups or test a hypothesis	What effect should the study be able to detect

Plan for **completed responses**. Expected nonresponse and clustered sampling can increase the number that must be contacted

The sampling frame and response process determine which population the completed survey represents

Sources: Lecture 5.1; Kim (2019), Section 3.5

An illustrative survey calculation

Suppose a study compares adoption rates of **50% and 60%** in two independent groups

With a two-sided 5% test and 80% power, a standard calculation calls for about **388 completed responses per group**

At an expected 80% response rate, that is about **485 invitations per group**. Clustering and ineligible contacts require further planning

Illustrative calculation using base R `power.prop.test(p1 = .50, p2 = .60, sig.level = .05, power = .80)`

Repeated tests create more chances for a false positive

If **20 independent null hypotheses** are each tested at the 5% level, the chance of at least one false positive is

$$1 - (1 - 0.05)^{20} \approx 0.64$$

The value changes when tests are dependent. A reader should look for the full set of tests, the planned hypotheses, and any adjustment for multiple comparisons

Source: Kim (2019), discussion of multiple testing

What changes with a large sample?

For a valid test with a fixed 5% level, the nominal Type I error rate remains **5%** under the null

Power rises for a specified nonzero effect. Estimates can become precise enough to detect effects too small to matter for the decision

Sampling, measurement, and model bias still require attention

Source: Kim (2019), Sections 2 and 3; statistical clarification in notes

Read the interval with the estimate

Suppose the estimated effect is **−2 units**, with a 95% interval from **−7 to 3**

The data are compatible with zero change and with a reduction of 5 units, which might matter for policy

A reviewer asks whether the interval rules out effects that matter, and whether the sample had adequate power for that question

When is an effect small enough to set aside?

An equivalence analysis starts by defining a substantively negligible range, such as $(-\Delta, \Delta)$

The research question determines Δ. Evidence for equivalence requires an interval sufficiently narrow to support an effect inside that range

Report the estimated effect and interval so a reader can apply a different decision threshold

Source: Kim (2019), Section 3.7

Practice: read this methods paragraph

We survey 300 farms in 12 irrigation districts before and after a revised water charge. Six districts adopt the charge. We estimate a difference-in-differences model of self-reported monthly water use, controlling for acreage and crop mix. Standard errors are clustered by district. The estimated change is −0.8 acre-feet, with a 95% interval from −2.2 to 0.6.

Illustrative classroom example

What would a reviewer ask next?

- How were farms and districts selected
- What supports the parallel-trends assumption
- How was monthly water use reported and checked
- Are 12 districts enough for the stated inference procedure
- What effect size would matter, and what can the interval rule out

Connect each question to a claim in the methods paragraph

Return to Tuesday’s methods draft

1. Underline the sentence that states your target quantity or question
2. Mark where you describe the sample, measures, comparison, and analysis
3. Add the assumption or design choice that needs the most explanation
4. State how you will report uncertainty, precision, or power

Use the workshop feedback to decide which part of the methods section to develop next

References

Falk, Armin, and James J. Heckman. 2009. “Lab Experiments Are a Major Source of Knowledge in the Social Sciences.” *Science* 326 (5952): 535–38. https://doi.org/10.1126/science.1168244.

Gibson, John. 2019. “Are You Estimating the Right Thing? An Editor Reflects.” *Applied Economic Perspectives and Policy* 41 (3): 329–50. https://doi.org/10.1093/aepp/ppz009.

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